

Autonomous battery management for mobile robots based on risk and gain assessment

Vincent Berenz · Fumihide Tanaka · Kenji Suzuki

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Abstract Battery management of mobile robots is an issue that has not been a strong focus of attention and is usually addressed by the simple use of battery thresholds. One of the main causes is that no significant method of assessment of risk of battery depletion has yet been proposed. As a result decision of redirection to a charging station is fixed and takes into account neither a dynamic evaluation of the risk of battery depletion nor an evaluation of the gain, defined as the level of mission accomplishment that could be achieved. In this paper we propose a novel method for evaluation of risk of battery depletion for mobile robots. Uncertainties concerning effective battery capacity, current discharge rate and energy required for reaching the station are addressed by the use of probability density functions. This risk assessment will allow replacing the usage of battery threshold by a customizable risk-taking parameter that will be used to define what level of gain is required for balancing a given level of risk. This risk/gain management of battery will guarantee that decision of redirection to the station corresponds to a favorable compromise between risk and level of mission accomplishment. While the proposed approach has been tested using a simulated and real room cleaning robot, it could be applied on a wider range of mobile robots.

Keywords Battery management · Mobile autonomous robots · Risk assessment · Energy aware operation

1 Introduction

For autonomous mobile robots capable of self-docking to a battery charging station, the standard solution for battery management is to set up a threshold of battery level at which the robot has to redirect itself to the docking station. This threshold value is not reevaluated depending on the current status of the robot, with the following drawbacks:

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- If the threshold is high enough to assure the safe return to the battery station for all situations, in many cases the robot will reach the charger with a surplus of remaining battery capacity that could have been used for longer operation time.
- If the threshold is low and cannot guarantee safe return to the docking station for all situations, there will be a risk for the robot not to be able to reach the docking station. The system will not ensure that this risk will correspond to significantly higher level of mission achievement.
- Battery thresholds are inflexible. They might order the robot to go back to the station at a critical time, when even a very short extension of operating time would have meant great benefit in terms of mission accomplishment.

These limitations derive from the fact that despite battery thresholds are used to avoid depletion of battery capacity, they do not rely on a measured risk of battery depletion: decision of redirection to the station depends neither on an evaluation of the risk of battery depletion nor on the level of mission accomplishment reached by the robot. If we define the gain as the level of mission accomplishment reached by the robot measured in percentage, possible hypothetical profiles for the progression of gain and risk with time is shown in Fig. 1. As the robot accomplishes its mission and uses energy from the battery, both gain and risk of the robot not reaching the docking station before battery depletion increase. Use of battery threshold will have the robot redirecting itself to the docking station at a time that does not attempt to optimize the gap between gain and risk.

The battery management proposed in this paper will express the problematic of redirection to the station as follow: considering expected future values of gain, what level of risk should

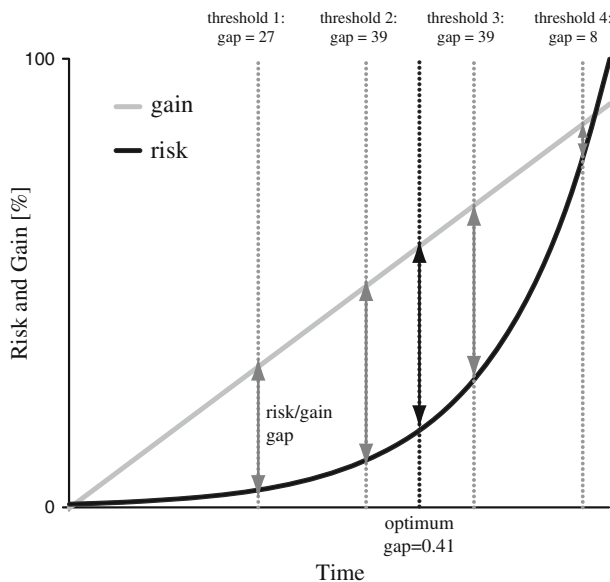


Fig. 1 Hypothetical profiles of increase of risk and gain against time, used for illustration purposes. Use of a battery threshold does not attempt to redirect the robot to the station at a time that maximizes the gap between gain and risk. Furthermore, similar modification in values of threshold may lead to uncontrolled variation in risk, gain, and corresponding gap: either a favorable variation (as from threshold 1 to threshold 2) or a drastically unfavorable variation (as from threshold 3 to threshold 4)

be considered acceptable? To implement such a management of battery, three requirements can be listed:

1. An evaluation of future increase of gain.
2. An evaluation of future risk of battery depletion.
3. An arbitrator deciding, with consideration for these two evaluations, when redirection to the station should be done.

This paper focuses on (2) and (3). An original assessment of risk of battery depletion for mobile robots based on probability density functions will be described, providing distinctive solutions based on whether or not the robotic system uses self localization and mapping technics.

For estimation of future gain, due to the intrinsically arbitrary definition of mission accomplishment, general solution cannot be provided: definition of gain will depend on the robotic system used, its particular task and how the user decided its evaluation. If the proposed method would be compatible with other gain evaluation functions, in this paper we will consider that gain can be approximated to a function linearly increasing with time. Such evaluation of future gain is very general and can be used by default for robots with low sensory ability for which other measure of level accomplishment can not be performed. The example of focus will be a commercially available room cleaning robot that moves randomly in the space in which it has to operate (Cohen et al. 2008). Consequently the proposed methodology will target having the robot considering risk taking only if resulting in a significantly longer operation time.

Using the proposed risk/gain battery management, the robot will decide when to redirect itself to the station in two steps. 1) It will determine if, in the time frame it could estimate future values of risk and gain, risk is always balanced by expected gain. What level of gain is required for balancing a given level of risk will depend on a configurable risk-taking parameter. This parameter will thus not refer to a fixed threshold of risk, but will rather indicate a tendency for the robot to be prone to risk-taking. 2) In the window of time corresponding to accepted levels of risk, a time corresponding to a large risk/gain gap will be selected as suitable for redirection to the station. Fixed decision of return to the station is thus replaced by an approach taking both assessed risk and mission accomplishment into consideration.

This paper is organized as follow. After a brief overview in Sect. 2 of research involving mobile robots and battery management, a methodology for risk assessment of battery depletion will be proposed in Sect. 3. Then, in the same section, we will describe how this risk assessment allows the determination of the best time at which the robot should redirect itself to the station. Finally, in Sect. 4, experiments using room cleaning robots will be used for both validating the proposed method of risk assessment and providing an example of the use of the proposed risk-taking parameter.

2 Related work

For autonomous systems, mission awareness can be considered as composed of three components: knowledge of mission objectives, internal self-health awareness and external self-situational awareness (Reichard 2004). Monitoring of the battery status can be considered as part of the self-health awareness which determines if the present system capability exceeds, meets or fails to meet mission demands. Reichard (2004) integrated health information in the form of battery level into an autonomous intelligent control system (AICS) which allowed

reallocation of mission responsibility to another member of a team of unmanned underwater vehicles when low capacity of power subsystem was detected. Another energy aware task reallocation was tested by Dressler and Fuchs (2005): estimation of remaining energy is done by observing decrease of battery voltage followed by linear regression and additional calibration. When a critical voltage value is reached, the robot initiates an emergency program and its current task is allocated to another robot. Lee et al. (2010) used a fuzzy logic decision making system for multiple task security robots that considered remaining battery energy as well as energy consumption of each possible task to control behaviors of robots.

Health awareness is a crucial aspect of Unmanned Aerial Vehicles (UAV). Roberts et al. (2008) tackled the energy problem of aerial exploration within indoor environments by using ceiling attachment as a mean for preserving energy and by providing a model for estimating the endurance of a hover-capable flying robot. The relationship between power consumption and thrust was determined, flight endurance being then estimated using take-off weight and battery capacity. To prevent damage, in case the battery voltage gets below a certain threshold, an automatic land feature is activated. Another research team examined the relationship between power usage, battery charge and other parameters in electric-powered quadrator flying vehicles and found a strong correlation between the battery voltage and the average collective stick position value (Bethke et al. 2010; Valenti et al. 2007). This was used to train a support vector regression model that could estimate non-invasively the hovering vehicles remaining flight time. The resulting estimated time was considered as health data that the Receding Horizon Task Assignment algorithm (Alighanbari 2004) was extended to consider as a constraint, insuring each vehicle to visit only waypoints that allow safe return to the station.

With the exception of the work from Roberts et al, in which threshold of battery for safe landing is used, remaining battery capacity is used for decision making or task attribution in a multi-robots context.

In no case the risk of battery depletion was numerically evaluated in a context where reallocation of the duty to another team member could not be performed.

3 Risk-gain management for autonomous robot

In this section, we will present a battery management system based on estimated future values of risk and gain. An overview of this risk/gain management strategy can be seen in Fig. 2.

We will first describe a methodology to assess the risk of a robot not reaching the docking station due to battery depletion. Such assessment of risk is difficult, as robotic mobile systems are characterized by uncertainties concerning battery capacity, energy required to go to the station and future discharge rates of the battery. The method we propose in this paper will address these uncertainties by the use of probability density functions.

It can be considered that, if a robot decides to go back to the docking station at a time t , it will successfully reach the docking station if its remaining battery capacity is higher than the energy required for navigating to the docking station. If uncertainties are introduced, both remaining and required energy are to be described by probability density functions (PDF), risk corresponding to their overlap:

$$R(t) = \int_{-\infty}^{+\infty} \left(\alpha(a, t) \times \int_a^{+\infty} \beta(b, t) db \right) da \quad (1)$$

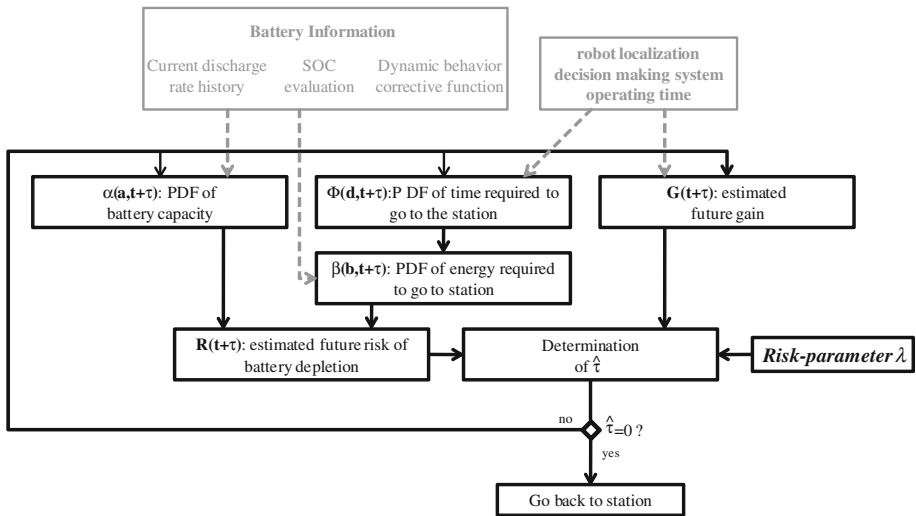


Fig. 2 Overview of risk-gain management of battery. At each operational time t , the robot generates functions returning expected values of gain and risk for future operational times τ . These functions are then used to determine $\hat{\tau}$, the operational time corresponding to high values of gap in the risk range allowed by the risk-parameter. If $\hat{\tau}$ corresponds to the current operational time ($\hat{\tau}$ is zero), the robot stops its current operation and redirects itself to the docking station

$\alpha(a, t)$ is the PDF of remaining actual capacity at time t , a referring to possible capacity values.

$\beta(b, t)$ is the PDF of required energy for navigating to the station, b referring to possible required energy values. It depends on the time required for the robot to navigate to the docking station and its average discharge rate.

The next sections will present methods for calculating $\alpha(a, t)$ and $\beta(b, t)$.

3.1 Estimation of remaining actual battery capacity

3.1.1 Generalities

If a battery is characterized by its ampere-hour capacity, in practice distinction must be made between the capacity inside the battery and the portion of this charge that will actually be available under the current discharge conditions. The charge effectively obtained by the battery will vary mostly with the charging/discharging rate, the temperature and the state-of-health (SOH) of the battery. The complex interrelationship of these factors causes difficulties in the pursuit of a precise SOC (State of Charge) estimation method. Many SOC estimation methods have been proposed in the literatures (Kikuoka et al. 1983; Seyfang 1990; Aylor et al. 1992; Huet 1998; Verbrugge 2002; Piller et al. 2001; Pei et al. 2006). General considerations can be found in (Pop et al. 2005) while solutions targeting specifically mobile robot are presented in (Lucas et al. 2005) and (Zhang et al. 2009). Discussing the relative benefits of the proposed solutions is not the purpose of this paper and depending on the system used, the most suitable methodology should be used.

For risk assessment, the corresponding estimation error E_a should also be evaluated. $\alpha(a, t)$ can be considered as a uniform PDF with values ranging from $C(t) - E_a$ and $C(t) + E_a$ with $C(t)$ being the SOC of the battery as evaluated at time t .

3.1.2 Battery dynamic behavior

In certain cases, for energy saving purposes, the robot might shut down certain services when navigating to the station, causing a lower discharge rate of the battery. This shut down of services will extend the operational time of the robot for two reasons:

1. Lower discharge rate implies that a given battery capacity will be delivered over a longer period of time.
2. The total capacity delivered by a battery depends on the discharge rate used. If the discharge rate is low, delivered capacity will be higher (Pop et al. 2005): the shut down of service will induce an higher effective battery capacity.

The second reason causes difficulties for the calculation of $\alpha(a, t)$: calculation prior to the shut down of services will produce an underestimation of the available capacity, as it does not take into account the future decrease in discharge current and the corresponding increase in battery effective capacity. But a priori incorporation of this supplementary capacity when estimating $\alpha(a, t)$ is problematic : the robot will not benefit from the supplementary capacity if the battery reaches its cut-off voltage before the services shut-down. In this paper we propose to incorporate this supplementary capacity in the calculation of $\alpha(a, t)$ and to apply a corrective function to the estimated risk. A practical example of such correction is given in Sect. 4.3.

3.2 Estimation of the time required to navigate back to the docking station

Imprecisions in measuring the duration time required for the robot to navigate to the charging station if it was to redirect itself to the docking station at time t might be due to uncertainties concerning the position of the robot relative to the docking station and the corresponding trajectory. Due to these imprecisions, we also treat the duration time required for navigating to the station as a PDF referred to as $\phi(d, t)$, d referring to possible values of duration. Two groups of robotic system can be distinguished for estimation of $\phi(d, t)$:

3.2.1 The robot does not use localization techniques

In such case the robot will only find the station after a phase of random search of uncertain length. This case can be commonly found in the field of home cleaning robot. In the example of Roomba (iRobot), the robot can detect the proximity of the station by detecting infrared signals emitted by it (Cohen et al. 2008). The robot can record the different times at which proximity to the docking station was detected and these data can be used to generate a PDF for estimating the frequency at which proximity to the docking station is expected. If at time t the robot detected the station k times, then the meeting of the station can be considered as a Poisson process of rate $\gamma = k/t$. Consequently, the PDF corresponding to the expected time between two events follows an exponential distribution characterized by a parameter of value γ .

3.2.2 The robot uses localization and mapping techniques along with path planning

At an operational time t , $\bar{d}$, the estimated time duration required for reaching the station, is to be obtained by the specific function implemented in the robot and uncertainties can be directly derived from the imprecision of the technique used. In such a case, $\bar{d}$ is associated with the corresponding error E_d . $\phi(d, t)$ can be considered as a uniform PDF for values of $\bar{d}$ ranging from $\bar{d} - E_d$ to $\bar{d} + E_d$.

3.3 Estimation of average discharge current during navigation to the docking station

Future average discharge rate when navigating to the docking station can be estimated based on the history of past discharge rates observed when the robot is operating under conditions similar to the one that will be used when navigating to the station. If robots activities never allow for such direct measurement, an history of estimations of such discharge rate have to be used. An example of such estimation will be given in Sect. 4.2. Uncertainties surrounding the estimation of future discharge rates depend on:

- Variations in discharge rate.
- The length of the period upon which the average of discharge rate has to be estimated: even if variations in discharge rate can be observed, if the period is long, high discharge rates are expected to compensate low discharge rates.

As future average of current discharge rates can not be estimated with complete certainty, the method proposed here will generate a PDF of possible expected averaged discharge currents $h(c, d, t)$, where c refers to possible values of averaged discharge current and d refers to the duration time over which the discharge current has to be averaged.

If the robot records its discharge rate at a frequency f and keeps n observations in memory, its history $Z(t)$ of discharge rate can be considered as a vector:

$$Z(t) = \{o_{t1}, \dots, o_{tn}\} \quad (2)$$

where o_{ti} is the discharge current as observed at previous operational time t_i .

$h(c, d, t)$ will be based on the vector $\bar{Z}_d$ of average of discharge rates over windows of width d :

$$\bar{Z}_d(t) = \{\bar{o}(0, d), \bar{o}(n - d, d)\} \quad (3)$$

with:

$$\bar{o}(i, d) = \frac{\sum_{j=i}^{f \times d + i} o_{tj}}{d} \quad (4)$$

$\bar{Z}_d(t)$ provides a sample of data representative of $h(c, d, t)$. The proposed methodology takes into account both:

- the uniformity of the environment: a more uniform environment will induce lower variations in values contained in $Z(t)$
- the length of the period upon which the average of discharge rate has to be estimated: longer period will induce lower variations in $\bar{Z}_d(t)$.

As a consequence the standard deviation of the resulting PDF will be expected to be higher for low values of d and lower for high values of d . Considering that $\bar{\phi}(d, t)$ provides

an estimation of the time duration required for the robot to navigate back to the station at time t (Sect. 3.2), $h(c, \bar{\phi}(d, t), t)$ is the PDF of averaged discharge current to be expected during this navigation.

3.4 Global risk assessment

3.4.1 General calculation

Estimations of time required to go back to the docking station and the average expected discharge rate can be used to generate $\beta(b, t)$, the PDF of energy required for going back to the docking station:

$$\beta(b, t) = \phi(d, t) \times h(c, \bar{\phi}(d, t), t) \quad (5)$$

The following simple Monte Carlo construction will allow the creation of a sample of values representative of $\beta(b, t)$:

- Generation of a vector ϕ consisting of random variables drawn from $\phi(d, t)$
- Generation of a vector H consisting of random variables drawn from $h(c, \bar{\phi}(d, t), t)$
- Generation of a new vector B obtained by multiplying ϕ and H element wise, constituting a random sample from $\beta(b, t)$

Finally, as expressed in equation (1), the risk of battery depletion $R(t)$ is the probability for an element randomly selected from $\alpha(a, t)$ to be inferior to an element also randomly selected from $\beta(b, t)$. In practice, a Monte Carlo construction similar to the one described above can be used for calculation of $R(t)$.

As the robot continuously updates the rate at which it detects the station and builds the history of past discharge rate, it creates more and more precise representations of $\alpha(a, t)$ and $\beta(b, t)$, and consequently produces better assessment of $R(t)$.

3.4.2 Battery dynamic behavior correction

As presented in Sect. 3.1, in case an increase of capacity due to the shut-down of services was introduced in the calculation of $\alpha(a, t)$, the calculation proposed above is expected to underestimate risk. $R(t)$ must be corrected by applying a corrective function. Experimental data are required to generate such a function (see Sect. 4.3.3).

3.5 Requirements for risk-gain management

We propose to replace the setting of battery threshold by a risk-taking parameter. As presented in the Sect. 1, a requirement for this approach is the ability for the robot to estimate future increase in gain. In this section we work under the assumptions:

- At an operational time t the robot can estimate its level of gain for the next τ^ϵ seconds: the function $g(t, \tau)$ is defined for $\tau \in [0, \tau^\epsilon]$.
- $g(t, \tau)$ never decreases with time

The strategy will consist of estimating future risk, estimating future gain and then determining the best time $\hat{\tau}$ at which the robot should redirect itself to the docking station.

3.6 Assessment of future level of risk, $R(t, \tau)$

Using the methodology for risk assessment proposed in the previous sections, the robot can assess $R(t)$, the present risk of battery depletion. Here we present estimation of future risks $R(t, \tau)$ for $\tau \in [0, \tau^\epsilon]$. To estimate $R(t, \tau)$ the following requirement must be met:

- Possibility for the robot to estimate energy required to navigate to the docking station at time $t + \tau$
- Possibility for the robot to estimate its remaining battery capacity at time $t + \tau$

3.6.1 Future energy required to navigate to the station

Equation (5) can be applied with estimation of $\phi(d, t + \tau)$ depending on the system used:

If the robot does not uses localization and path planning The time required to go the docking station can not be estimated based on the position of the robot, always unknown, and $\phi(d, t + \tau) = \phi(d, t)$.

If the robot uses localization If the robot is navigating to a determined position, future positions relative to the station may be estimated along with the time required to navigate to it. If future positions can not be estimated: $\phi(d, t + \tau)$ will be a uniform PDF with values ranging from $(D(t) - v \times \tau)/v$ to $(D(t) + v \times \tau)/v$ where $D(t)$ corresponds to the current distance to the station and v corresponds to the maximum speed of the robot.

3.6.2 Future remaining battery capacity

$\alpha(a, t + \tau)$, PDF of estimated actual battery capacity at time $t + \tau$, can be estimated from $\alpha(a, t)$ and the history of past discharge current as presented in Sect. 3.3:

$$\alpha(a, t + \tau) = \alpha(a, t) - \tau \times \bar{h}(c, \tau, t) \quad (6)$$

Estimation of $R(t + \tau)$ follows, using the Monte Carlo construction described in Sect. 3.4.

3.7 Strategic time determination

Once both $g(t, \tau)$ and $R(t, \tau)$ are estimated for $\tau \in [0, \tau^\epsilon]$, the robot can specifically choose to navigate back at a time $\hat{\tau}$ depending on a configurable risk-taking parameter which value is within $[0, 1]$. A low value of risk-taking parameter will give a tendency to avoiding risk, while a high value will authorize the robot to take higher risk, but only if balanced by higher gain. We propose the following strategy for the robot to select the time $\hat{\tau}$ at which it should redirect itself to the docking station, as presented in Fig. 3:

- Normalization of $g(t, \tau)$, the resulting function being $G(t, \tau)$
- Defining $\tau^{max}(\lambda)$ as the time such as $\tau^{max}(\lambda) \leq \tau^\epsilon$ and $\forall \tau \in [0, \tau^{max}(\lambda)]$:

$$G(t, \tau) > \left(\frac{1}{\lambda} - 1 \right) R(t, \tau) \quad (7)$$

- If $R(\tau^{max}(\lambda)) < 1.0$, then $\hat{\tau} = \tau^{max}(\lambda)$. Otherwise $\hat{\tau}$ is the time τ in $[0, \tau^{max}(\lambda)]$ for which a score $S(\tau)$ is maximal with:

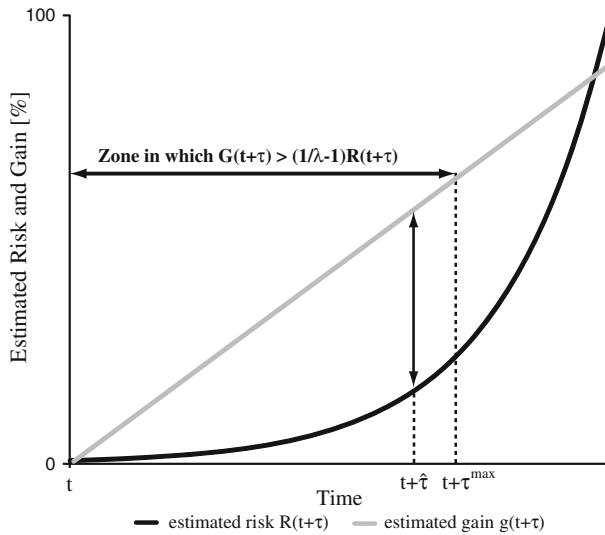


Fig. 3 Overview of strategic time determination at time t . $[t, t + \tau^{\max}]$ corresponds to the interval of future time in which, according to the chosen risk parameter λ , risk taken is justified in regard to the estimated gain. $t + \hat{\tau}$ represents a time of compromise between high gain and high gap between gain and risk in $[t, t + \tau^{\max}]$

$$S(\tau, \lambda) = \frac{G(t + \tau)}{G(t + \tau^{\max}(\lambda))} (G(t + \tau) - R(t + \tau)) \quad (8)$$

The value $\tau^{\max}(\lambda)$ guarantees that for all value of $\tau \in [0, \tau^{\max}(\lambda)]$, risk taking will be balanced by an increase of gain considered significant in regard to the risk-taking parameter of the robot.

Low values of risk-taking parameter will induce stricter requirement in terms of gain than high values of risk-taking parameter, and the robot will be less prone to take risk. In the range $[0, \tau^{\max}(\lambda)]$, $\hat{\tau}$ corresponds to the maximal of a score representing a compromise between higher gain and high gap between gain and risk.

As $S(\hat{\tau}, \lambda)$ corresponds to a maximum in the interval $[0, \tau^{\epsilon}]$, the importance of τ^{ϵ} can be underlined. If the robot can estimate future values of gain over a larger window of time, it will be more likely to redirect itself to the station at a time $\hat{\tau}$ corresponding to an absolute maximum of $S(\tau, \lambda)$.

3.8 Setting the risk-taking parameter

As expressed in Eq. (7), if λ is equal to zero, the required gain value for taking any risk is infinite: the robot will redirect itself to the station before taking any risk of battery depletion. Thus setting such a value of λ guarantee safe docking. On the other hand, if λ is equal to 1, the robot will be allowed to take 100% of risk for any expected value of gain. This value of λ expresses that the robot should always pursue increase of gain, even in case of very low battery level. The difference with the use of a very low battery threshold is that, if no increase of gain is to be expected, the robot will redirect to the station. Between these two extremes, increasing values of λ will imply lower requirement in terms of gain/risk ratio for the robot to keep operating in situation of lowered battery capacity.

4 Experiments

4.1 Objectives and experimental setup

Experiments were performed with three objectives:

1. Providing a concrete example of implementation of the proposed risk/gain management of battery. This example is in the field of room cleaning robot.
2. Validating the proposed method for assessment of risk by conducting sets of experiments. For each of these sets, the robot was required to redirect itself to the station when the assessed risk reached a certain value that was used as a threshold of risk. If the proposed method for risk assessment is correct, rate in failure to reach the station before battery depletion should correspond to the threshold of risk used.
3. Analyzing the results in terms of risk and gain obtained for different values of risk parameter when the proposed methodology is used. A set of experiments was conducted and values of risk and gain corresponding to different values of risk-parameter and battery threshold were recorded.

For (2) and (3), a very large number of experiments is required in order to obtain statistically relevant results. Because of this, building a reliable simulation was a strong requirement. A simulation was built and used for:

- Performing the very high number of experiments required to obtain statistically significant results
- Measuring the effect of a battery dynamic behavior resulting from a shut down of services and generating a corresponding corrective function as mentioned in Sect. 3.4.2.

Considering the points above, experiments were conducted as follow:

- A first set of 25 pilot experiments were conducted using a real robot. Data extracted from these experiments were used for: 1) calibrating a SOC estimation methodology based on coulometry, 2) measuring the error in SOC estimation, 3) estimating the effect of battery dynamic behavior and 4) extracting real robot values of current consumptions required for building a reliable simulation.
- Following these experiments, a simulation was built based on real robot data and 6,000 experiments were conducted for validating the proposed risk assessment, evaluating the effect of battery dynamic behavior and generating a corresponding corrective function.
- Risk assessment using this corrective function was then evaluated in the real robot through 80 experiments.
- Finally, 200 supplementary experiments were conducted in the simulation in order to analyze risk and gain values corresponding to different values of risk parameter.

All real robot experiments were performed using a Roomba room cleaning robot (version 577, iRobot) equipped with a NiMH 14.4 V 2,000 mA h battery for which the cut off voltage was set at 13.5 V. The robot was remotely controlled through a bluetooth connection and the ROI (Roomba Open Interface). A minimal controller making the robot moving forward and turning a random angle between 90 and 270 degrees in case of collision was used. When navigating back to the station, the cleaning motors were turned off for energy saving purposes, the robot docking to the station when finding itself at its proximity. The shut down of cleaning services made the robot subject to sudden increase of battery effective capacity due to dynamic behavior of the battery. The robot operated in a room of 17.5 m², with variations concerning parameters as presented in Table 1. The temperature was not controlled but variation in the battery block temperature were observed, depending mostly on the interval of time

Table 1 parameters varying across experiments

Parameter	Range
Pieces of furniture	Between 0 and 5 (max width of 100 cm)
Floor type	Carpet or vinyl
Robot speed	Between 0.5 and 2 cm s ⁻¹
Battery block temperature	Between 20 and 50°C (not controlled)

between charge and discharge. The original docking station was placed in an open box during all experiments. This was done in order to limit the area in which the robot would detect the emitted infrared signal and thus emulate a room of bigger size. The infrared signal could be detected by the robot in a area of 42 × 42 cm. Variations in current consumption during operational time were observed, ranging from 1,300 to 1,700 mA, lower current consumption being observed when the robot is turning.

4.2 Real robot: pilot experiments

The first set of experiment in the real robot was performed to gather data required for both applying the proposed risk assessment methodology and building a reliable simulation.

4.2.1 Estimation of actual battery remaining capacity

First, simple coulomb counting was used for estimation of battery remaining capacity under the assumption of a fully charged 2,000 mAh battery. Experimental data over 15 full cycles showed that assumption of a 2,000 mAh effective capacity would lead to an estimation error up to 12.5%: observed effective capacity for the 15 experiments are shown in black in Fig. 4. Corrective functions using the average temperature at the battery block, the averaged discharge current and the initial battery voltage as parameters were generated using linear regression. As the battery used was new (number of cycles under 120 for all experiments), effects of SOH was neglected. One in one out cross validation of estimation of battery capacity using the corrective function was used: estimated effective capacities are represented in grey in Fig. 4 and shows that the error lowered to 5.4%. The resulting $\alpha(a, t)$ is a uniform PDF of values ranging from C_{min} to C_{max} , with

$$C_{min} = C - C \times 0.054 \quad (9)$$

and

$$C_{max} = C + C \times 0.054 \quad (10)$$

C refers to the battery capacity as estimated using the corrective function.

4.2.2 Battery dynamic behavior

As the robot turns off its cleaning motors when redirecting itself to the station, the discharge rate of the battery is suddenly lowered, inducing an increase in expected battery capacity (Sect. 3.1.2). This increase of capacity is made explicit in Fig. 5, which shows variations of battery voltage for two experiments for which the observed initial voltage (16.754 V), the average battery temperature (46.8 and 44.5°C) and the average discharge

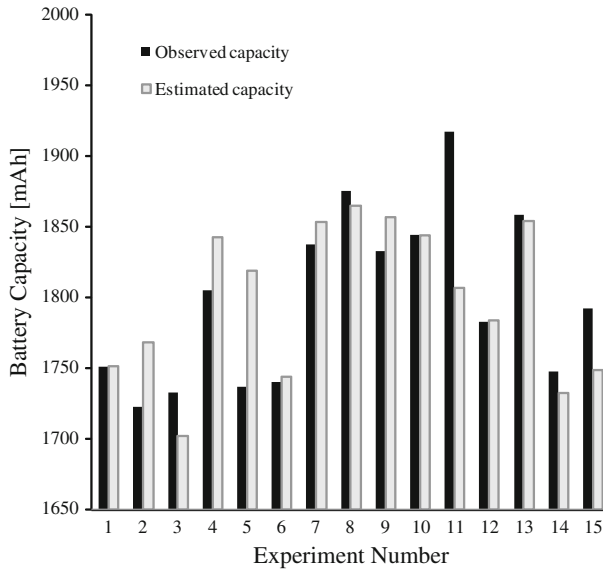


Fig. 4 Observed effective battery capacity and estimated effective battery capacity for 15 battery cycles. Estimation was based on using a corrective score depending on initial battery voltage, average current consumption and temperature of the battery block

current (1,525 and 1,527 mA) were almost identical. For these conditions, application of the methodology exposed above indicates a battery effective capacity comprised between 1,689 and 1,883 mA h. For experiment 1, total capacity delivered over the experiment was 1,756 mA, which is in the range of expected results. For experiment 2, for which cleaning motors were turned off at time 4,100 s, the final capacity was 1,896 mA h, which is above the range of expected capacities. When the motors are turned off, resulting in the discharge current suddenly decreasing from 1,500 to 550 mA h, a sharp increase of voltage can be observed. Linear regression (dotted line in Fig. 5) was used to estimate that if the cleaning motors had not been turned off, the end of operation time would have been 4,327 s which would have corresponded to a delivered capacity of 1,837 mA h. It can thus be considered that the robot gained 59 mA h due to the sudden decrease of current discharge rate, which corresponds to more than 6 min of battery life under non-cleaning navigation conditions.

Pilot experiments using assessed risk thresholds of 30 and 70% illustrated the consequences of this increase in capacity. While failure to reach the station due to battery depletion was observed when using a threshold of assessed risk of 70% (2 failures for 3 battery cycles), no failure was observed for 10 experiments using a risk threshold of 30%: the unexpected gain in battery capacity due to shut down of cleaning motors always allowed safe docking. An assessment of risk not taking the effect into account produces an overestimation for lower values of risk.

Experimental data collected over 10 cycles for which the robot would turn off its cleaning motors at different operational times were used to quantify this increase in capacity. These data showed an increase of battery capacity between 1.6 and 5.4%. This supplementary capacity is to be included in $\alpha(a, t)$ by using corrected values of C_{min} and C_{max} for the generation of the corresponding uniform PDF:

$$C_{min}^{corrected} = C_{min} + C_{min} \times 0.016 \quad (11)$$

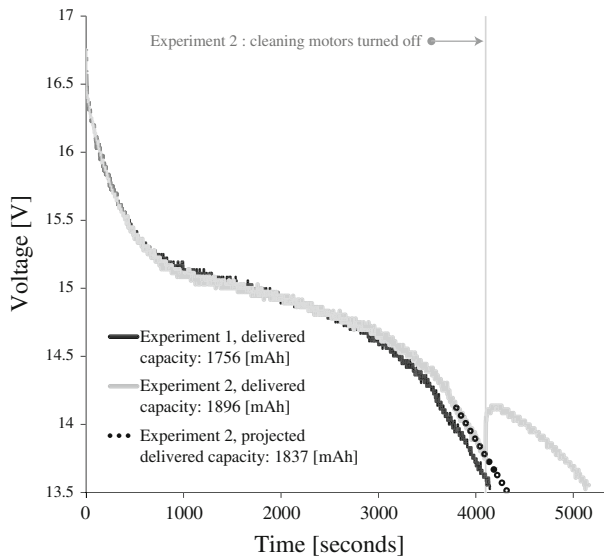


Fig. 5 Battery dynamic behavior. Both experiments were conducted under the same conditions in terms of temperature, current discharge and initial voltage. For experiment 2, at time $t = 4,100$ s, cleaning motors of the robot were turned off, resulting in a gain of battery capacity of 59 mAh

and

$$C_{max}^{corrected} = C_{max} + C_{max} \times 0.054 \quad (12)$$

4.2.3 Estimation of discharge current during navigation to the docking station

As during operation the robot has cleaning motors constantly turned on, no direct measurement of the discharge current required for navigation with cleaning motors turned off is available. Experimental data showed that simple linear regression over observed global current discharge and observed current consumption of cleaning motors allows its precise estimation (R^2 of 0.997), rendering possible the generation of the history of current required for navigation only as described in 3.3.

4.2.4 Estimation of the time required to go to the docking station

During operation, the robot can detect proximity to the docking station by detecting the IR signals it emits. As presented in Sect. 3.2.1, the robot records the times at which the docking station is detected and uses these data to generate the PDF representing the interval of time at which proximity to the docking station should be expected.

4.2.5 Simulation building

A simulation of the robot and its environment was built. Current consumption was modeled using real robot data: depending on activity of the virtual robot (moving straight or turning), current consumption values were randomly selected from a corresponding set of data containing current consumption rates observed in the real robot. Realistic variations in

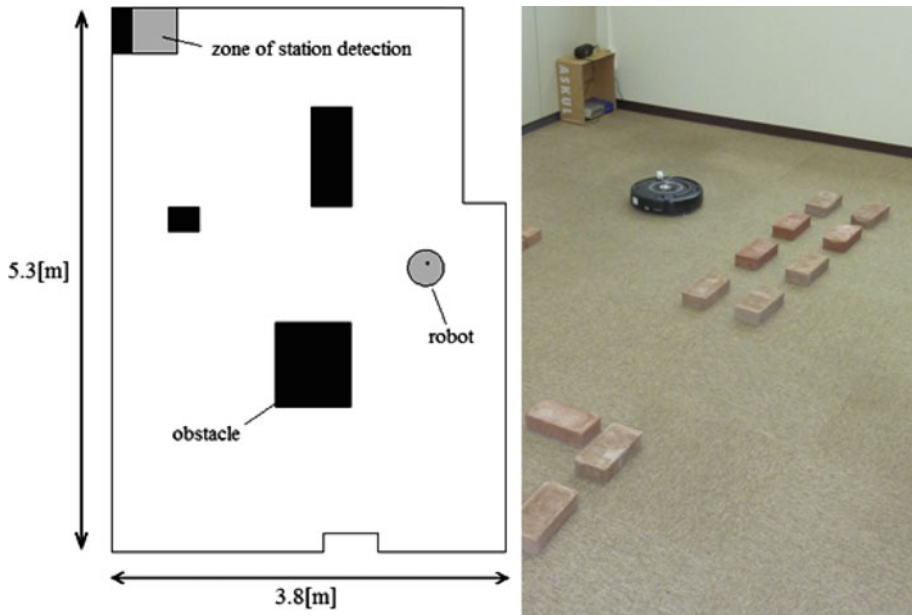


Fig. 6 Robot experiment. *Left* Simulated room and robot. *Right* Real robot experiment

discharge rate were thus implemented. Uncertainties in available battery capacity estimation were introduced by randomly selecting for each experiment a capacity in the error range observed in the real robot. The controller implemented in the simulated robot was the same as the one used in the real robot and the simulation was programmed to generate automatically diverse situations in terms of furnitures, temperature, flooring (vinyl or carpet) and robot speed. Frequencies at which the robot would detect the station in both the simulation and real environments were observed to be similar under same conditions (data not shown). A representation of the simulated environment and a picture of the real robot environment can be seen in Fig. 6.

4.3 Simulation experimental results

4.3.1 No battery dynamic effect

A first set of virtual experiments was done in order to validate the proposed assessment of risk. For this set of experiments, the robot was not required to turn off its cleaning motors when navigating back to the station and as a consequence, no battery dynamic effect was to be observed: Eqs. (11) and (12) were not used for the calculation of $\alpha(a, t)$. A random set of 200 environments was generated and for each of them, 10 experiments with different values of risk thresholds above which the robot was required to return to the station were performed. For each threshold of risk, the number of times the robot would fail to dock to the station was recorded. Results are shown as dotted black line in Fig. 7: failure rates correspond to the risk thresholds used (R^2 of 0.996). These results validate the PDF based approach of risk assessment. As variations of discharge current and temperature were modeled based on real robot data, and as frequencies at which the robot would meet the station were observed to be

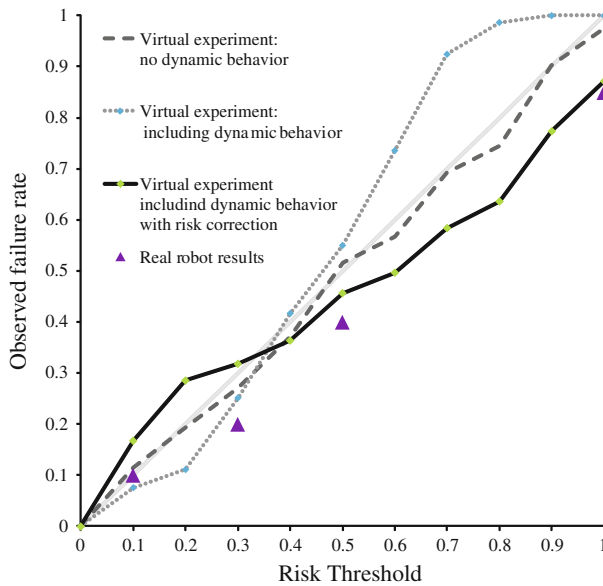


Fig. 7 Observed failure rates for different values of assessed risk thresholds in virtual and real robot experiments. 200 experiments were conducted for each risk threshold in virtual experiments, 20 experiments were conducted for each risk threshold in real robot experiments

similar for the real and the simulated robot, it can be considered that eventual error in risk assessment when using the real robot would have for origin:

- An error in estimation of the error related to the SOC evaluation method. In the simulation, if it was not assumed that the robot could estimate perfectly the delivered capacity, which was randomly generated in the error range observed in real robot, it was assumed that the error surrounding this estimation was perfectly estimated.
- A non negligible battery dynamic effect due to the shut down of cleaning motors when navigating back to the station.

The next set of virtual experiments was conducted in order to measure the effect on risk assessment induced by having the cleaning motors turned off when navigating to the station.

4.3.2 Battery dynamic behavior

In a second set of virtual experiments, the robot was required to turn off its cleaning motors when navigating to the station, thus becoming subject to battery dynamic behavior. The simulation was updated to take the effect in consideration and $\alpha(a, t)$ was generated taking into account the supplementary delivered capacity: Eqs. (11) and (12) were used. A new set of experiments was conducted, results being shown by the small dotted line in Fig. 7: if correlation between risk threshold and failure rate remains high (R^2 of 0.960), it can be observed that risk is systematically underestimated for high value of threshold. This underestimation is explained by the fact that for higher level of risk, the robot sometimes gets its battery depleted before turning its cleaning motors off: in such cases, it does not benefit from the extra capacity that has been included in the calculation of $\alpha(a, t)$ in expectation of the decrease of discharge current. As a consequence, a correction of the assessed risk is required.

4.3.3 Battery dynamic behavior and risk correction

The curve obtained in the previous set of experiments was approximated to a five degree polynomial function that was used for correcting the estimation of risk. Results of a new set of experiments using corrected risk estimation are shown as a plain black line in Fig. 7. They show higher correlation (R^2 of 0.978) and the underestimation of risk for high value of threshold is not to be observed anymore.

4.4 Real robot experiments

The methodology, using the corrected assessed risk, was tested in the real robot for four risk thresholds: 10, 30, 50 and 100%. The exponential increase of risk (see Sect. 4.5 below) could not allow testing for thresholds comprised between 50 and 100%: increase from 50% of risk to 100% would happen in <30 s, interval of time for which the procedure could not be precise enough as sampling for the use of the Monte Carlo procedure required to use averaged values of risk over 10 s. For each tested threshold, 20 experiments were conducted. Results are also displayed in Fig. 7 and show a reasonable overestimation of risk. As determined in the first set of virtual experiments (Sect. 4.3.1), this overestimation derives either from a misrepresentation in the error surrounding the battery capacity estimation or from imprecisions due to the use of the dynamic effect corrective function. As a linear increase of failure rates can be observed, use of the corrective function can be considered successful.

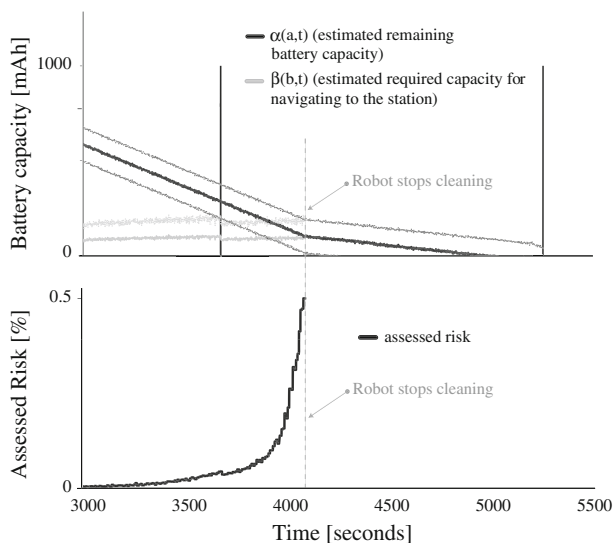


Fig. 8 Evolution with time of $\alpha(a, t)$, $\beta(b, t)$ and the corresponding assessed risk for a real robot experiment for which the risk threshold was set to 50%. Horizontal axis represents time. *Top* PDF corresponding to $\alpha(a, t)$ and $\beta(b, t)$ are represented by their average and standard deviation. *Vertical bars* indicate times at which the station was detected. *Bottom* corresponding assessed risk. For time comprised between 0 and 3,000 s, the assessed risk was always equal to zero. At time $t = 4, 096$ s, the risk threshold is attained and the robot turns off its cleaning motors. At time $t = 5, 273$ s the robot detects the docking station and the experiment stops

4.5 Risk profile

Figure 8 shows the evolution with time of $\alpha(a, t)$, $\beta(b, t)$ and the corresponding assessed risk for a real robot experiment for which the risk threshold was set to 50%. PDF corresponding to $\alpha(a, t)$ and $\beta(b, t)$ are represented by their average and standard deviation. It can be seen that risk increases as the PDF corresponding to $\alpha(a, t)$ and $\beta(b, t)$ overlap. At time $t = 4,096$ s, the risk threshold is attained and the robot turns off its cleaning motors. Resulting decrease in discharge current induce a change of the rate the remaining battery capacity decreases. As decision of redirection to the station has been taken, calculation of $\beta(b, t)$ is stopped. Shortly after, as the PDF of $\alpha(a, t)$ starts including non-positive values, battery depletion could happen anytime. As for this particular run the battery effective capacity turned out to be in the higher range of expected capacities values, the robot could operate until time $t = 5,273$ s, time at which it detected the docking station, and the experiment stopped. An exponential rise of risk can be observed and explained by the combined effects:

- Averaged discharge current when cleaning is 1,522 mA, while being 539 mA when navigating with the cleaning motors turned off. This difference in discharge current roughly means that if the robot cleans 1 min more, it will have 3 min less for finding the station.
- As time goes by, the following two risks accumulate : 1) risk of reaching the voltage cut off before turning the cleaning motors off and thus not benefiting from the extra capacity resulted from the sudden decrease in current discharge rate, 2) risk of not reaching the station even in case the cleaning motors have been turned off before reaching the voltage cut off.

Such a risk profile justifies the use of a risk/gain management of battery: risk of battery depletion remains very low for a long period of time followed by a sudden sharp increase. As it will be showed in the next section, estimation of the moment this sharp increase of risk will happen can allow the robot to stop operating right before its occurrence.

4.6 Evaluation of risk-gain management

The proposed method of risk-gain battery management was used to extend the service domain of cleaning robots. It is usually recommended to use such robots in areas in which the robot has sufficient battery autonomy for safely finishing its cleaning task. Here we propose the use of cleaning robots in areas for which the battery capacity is not necessarily high enough to complete cleaning in one cycle. In such cases, benefit-risk management is important as risk taking must be carefully considered:

- allowing the robot to take some risk might extend its operating time, thus reducing in the long term the frequency at which it will need to dock to the station.
- if the robot runs out of battery before reaching the docking station, it will not be able to recharge and perform another cycle of cleaning.

The simulation was used to compare results obtained in large rooms for different values of risk-taking parameters and battery thresholds. Environments were generated randomly as previously described, but would be of a surface comprised between 100 and 200 m², size for which full cleaning can not be performed on a single charge. The drop in the rate at which the robot would detect the station is compensated by adding a second docking station. As the objective was to have the robot cleaning as long as possible, we selected a gain function corresponding to the operational time of the current cleaning cycle:

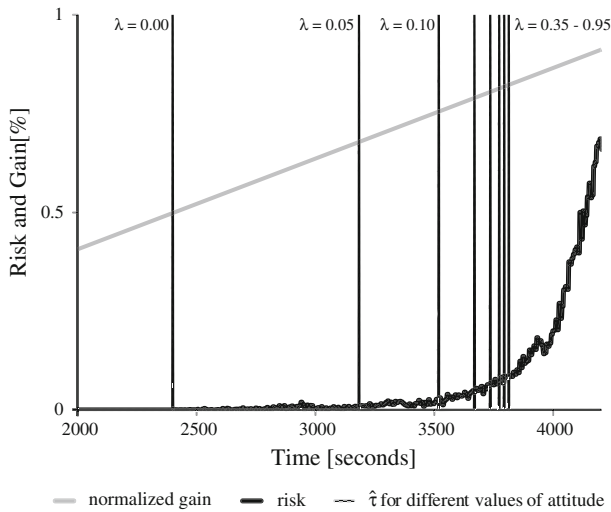


Fig. 9 $\hat{t}$ for values of risk-taking parameter comprised between 0.0 and 0.95, with an increase step of 0.05

$$G(t) = \frac{t}{t^{max}} \quad (13)$$

where t^{max} corresponds to the longest time the robot could operate, as estimated using the averaged discharged current and the estimated battery capacity.

Two hundred experiments were conducted. For each experiment, the robot was required to operate until battery depletion. Times at which it would have redirected itself to the station, along with corresponding values of gain and assessed risk, were recorded for different values of risk-taking parameter and battery threshold.

Figure 9 presents, for a typical experiment, return policies for different values of risk-taking parameter. Evolution of risk and gain with time are represented respectively in dark and light gray, vertical bars representing times at which the robot would redirect itself to the station for different values of risk-taking parameter. For a risk-taking parameter of 0.0, the robot would seek for the docking station as soon as a non negligible risk is detected. For higher values of risk-taking parameter, the robot considers that risk taking is justified in regard to the increase in gain and it decides to operate longer. Values of risk-taking parameter comprised between 0.35 and 0.95 refer to the same value of $\hat{t}$: even for high level of risk-taking parameter, the robot avoids the time zone corresponding to the sharp increase of risk.

Figure 10 summarizes the results obtained on the 200 experiments. Values of gap obtained for different values of both battery thresholds and risk-taking parameters are displayed. Corresponding average values of risk and gain are also represented. It can be seen that variation of thresholds may lead to uncontrolled variation in gap. On the other hand, use of the proposed risk-taking management has the robot targeting the best compromised of gain and risk. Furthermore, for no value of risk-taking parameter, the robot would take a risk that is not compensated by significantly higher value of gain and consequently the robot would always avoid the sharp exponential increase of risk.

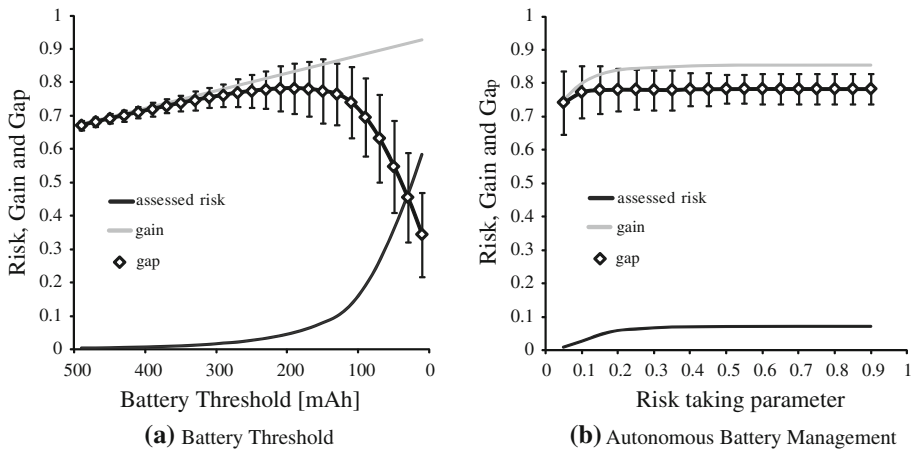


Fig. 10 Difference between gain and risk (gap) at $\hat{\tau}$ for different values of battery thresholds and risk-taking parameters, along with corresponding standard deviations. Results collected over 200 virtual experiments

5 Conclusion

This paper introduced the idea that battery management of mobile robot should not consist in simple avoidance of battery depletion: in certain situations, risk taking is justified in regard of the level of mission accomplishment that can be attained. The proposed method of risk/gain management allowed a cleaning robot to determine that a risk level of 10% would result in an increase of more than 15 min of operating time (Fig. 9). As the robot could also determine that higher levels of risk would not lead to such significant increases of operating time, it would systematically redirect itself to the station at a time showing favorable compromise between gain and risk.

The gain function used in the experimental part simply referred to the operation time. If this general gain function could be used as default for a broad range of applications, the proposed method would be applicable to other measures of level accomplishment. A salient example would be a step function to be used for robots which duty targets defined items: when such an item is detected, a step in gain is to be expected. If no such item is detected, no step in gain is to be expected. The robot can thus estimate future steps in gain at least a small window of time ahead. An example of such a robot would be DustBot (Ferri et al. 2010): this trash collecting robot operates in the pedestrian area of urban environments and comes to collect garbage bags on demand of citizens, at their doors. As DustBot uses preloaded area map for navigation, it can estimate the navigation time required to go to the location of bag picking. When receiving a demand, it can thus estimate the approximate time at which a step in gain would happen. The proposed approach would allow the robot, when receiving a demand, to decide between redirecting itself to the station before or after collecting the trash.

Future work will include evaluating the proposed methodology when applied to other robotic systems and to a larger range of gain functions.

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